

Taller 2

Curso: Matemática Estructural, Carla Gonzalez Mina

Due: Lunes 31 Agosto, 2026

Exercise 1.

Sea n un número entero. Demuestra que si n es impar, entonces n^3 es impar.

Exercise 2.

Sean a y b dos números racionales tales que $a < b$. Demuestre que el promedio ponderado $\frac{1}{3}a + \frac{2}{3}b$ es un número racional entre a y b .

Exercise 3.

10 P.

Show that a 2×2 symmetric real matrix always has real eigenvalues.

Exercise 4.

10 P.

Let V be the vector space of polynomials of degree at most 2. Show that $\{1, x, x^2\}$ is a basis for V .

Answer Key

Solution 1.

The characteristic polynomial is $\det(A - \lambda I) = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3$, which factors as $(\lambda - 1)(\lambda - 3)$. So the eigenvalues are $\lambda = 1$ and $\lambda = 3$.

Solution 2.

For $\lambda = 1$: $(A - I)v = 0$ gives $v = (1, -1)^\top$ (up to scale). For $\lambda = 3$: $(A - 3I)v = 0$ gives $v = (1, 1)^\top$ (up to scale).

Solution 3.

For $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ the characteristic polynomial is $\lambda^2 - (a + c)\lambda + (ac - b^2)$, with discriminant $(a + c)^2 - 4(ac - b^2) = (a - c)^2 + 4b^2 \geq 0$. A non-negative discriminant means both roots are real.

Solution 4.

Spanning: any $p(x) = a_0 + a_1x + a_2x^2$ with $a_i \in \mathbb{R}$ is by definition a linear combination of $1, x, x^2$.

Independence: if $a_0 + a_1x + a_2x^2 = 0$ for all x , comparing coefficients forces $a_0 = a_1 = a_2 = 0$. So the three polynomials are linearly independent, and together they form a basis.